

EX.NO:13 Rootkit Detection and Analysis in Linux Using Rkhunter

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Rootkits are a type of malicious software designed to gain unauthorized access to a system while remaining hidden from detection. They often modify system files, processes, or kernel modules to maintain persistence and avoid being discovered by traditional security tools. Detecting such threats requires specialized utilities that can analyze system integrity and identify unusual behavior.

Tools like Rkhunter are commonly used to scan for known rootkits, suspicious configurations, and unexpected changes in critical system files. Rkhunter works by comparing current system states with trusted values and looking for anomalies. However, since no single tool can guarantee complete detection, additional utilities such as *unhide* and *lsmod* are used to crossverify results.

This experiment focuses on analyzing a system for potential rootkits and suspicious modifications, while emphasizing the importance of using multiple tools to improve detection accuracy and reduce false positives.

Objective

To set up and utilize Rkhunter for identifying rootkits and unusual system

activity To examine system integrity using a combination of security tools

To evaluate and compare the detection outcomes produced by different utilities

Tools Used

- Rkhunter
- unhide
- lsmod, strings
- Linux

Procedure

Task 1 - System Preparation and Baseline

1. Creating Snapshot
2. Install the tool Rkhunter

```
remnux@remnux: ~  
remnux@remnux:~$ sudo apt install rkhunter  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  bsd-mailx liblockfile-bin liblockfile1 libnsl2 postfix ssl-cert unhide.rb  
Suggested packages:  
  postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver postfix-mysql  
  postfix-pcre postfix-pgsql postfix-sqlite procmail sasl2-bin | dovecot-common ufw powermgmt-base  
The following NEW packages will be installed:  
  bsd-mailx liblockfile-bin liblockfile1 libnsl2 postfix rkhunter ssl-cert unhide.rb  
0 upgraded, 8 newly installed, 0 to remove and 281 not upgraded.  
Need to get 1627 kB of archives.  
After this operation, 5772 kB of additional disk space will be used.  
Do you want to continue? [Y/n] Y  
Get:1 http://us.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]  
Get:2 http://us.archive.ubuntu.com/ubuntu noble/main amd64 libnsl2 amd64 1.3.0-3build3 [41.4 kB]  
Get:3 http://us.archive.ubuntu.com/ubuntu noble/main amd64 postfix amd64 3.8.6-1build2 [1254 kB]  
Get:4 http://us.archive.ubuntu.com/ubuntu noble/universe amd64 rkhunter all 1.4.6-12 [219 kB]  
Get:5 http://us.archive.ubuntu.com/ubuntu noble/main amd64 liblockfile-bin amd64 1.1.2-1build1 [11.3 kB]  
remnux@remnux: ~
```

3. Verify the Installation

```
remnux@remnux: ~  
remnux@remnux:~$ rkhunter --versionm  
Invalid option specified: --versionm  
remnux@remnux:~$ rkhunter --version  
Rootkit Hunter 1.4.6  
  
This software was developed by the Rootkit Hunter project team.  
Please review your rkhunter configuration files before using.  
Please review the documentation before posting bug reports or questions.  
To report bugs, provide patches or comments, please go to:  
http://rkhunter.sourceforge.net  
  
To ask questions about rkhunter, please use the rkhunter-users mailing list.  
Note this is a moderated list: please subscribe before posting.  
  
Rootkit Hunter comes with ABSOLUTELY NO WARRANTY.  
This is free software, and you are welcome to redistribute it under the  
terms of the GNU General Public License. See the LICENSE file for details.
```

4. Establish the Baseline

```
Rootkit Hunter 1.4.6

This software was developed by the Rootkit Hunter project team.
Please review your rkhunter configuration files before using.
Please review the documentation before posting bug reports or questions.
To report bugs, provide patches or comments, please go to:
http://rkhunter.sourceforge.net

To ask questions about rkhunter, please use the rkhunter-users mailing list.
Note this is a moderated list: please subscribe before posting.

Rootkit Hunter comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it under the
terms of the GNU General Public License. See the LICENSE file for details.

remnux@remnux:~$ sudo rkhunter --propupd
[ Rootkit Hunter version 1.4.6 ]
File updated: searched for 180 files, found 139
remnux@remnux:~$
```

Purpose:

- Maintains hash values of important system files
- Utilized later to verify integrity by comparing these hashes

Task 3 - Perform System Scan

```
remnux@remnux:~$ sudo rkhunter --check
[ Rootkit Hunter version 1.4.6 ]

Checking system commands...

Performing 'strings' command checks
```

Task 4 - Analyze the Scanned Results

1. Summary Output

```
remnux@remnux:~$ sudo rkhunter --check
[ Rootkit Hunter version 1.4.6 ]
```

```
Checking system commands...
```

```
Performing 'strings' command checks
  Checking 'strings' command                [ OK ]

Performing 'shared libraries' checks
  Checking for preloading variables         [ None found ]
  Checking for preloaded libraries         [ None found ]
  Checking LD_LIBRARY_PATH variable        [ Not found ]

Performing file properties checks
  Checking for prerequisites                [ OK ]
  /usr/sbin/adduser                        [ OK ]
  /usr/sbin/chroot                         [ OK ]
  /usr/sbin/depmod                         [ OK ]
  /usr/sbin/fsck                          [ OK ]
```

2. To View Detailed Logsudo less /var/log/rkhunter.log

```
remnux@remnux: ~
[12:36:58] Info: Found the 'basename' command: /usr/bin/basename
[12:36:58] Info: Found the 'diff' command: /usr/bin/diff
[12:36:58] Info: Found the 'dirname' command: /usr/bin/dirname
[12:36:58] Info: Found the 'file' command: /usr/bin/file
[12:36:58] Info: Found the 'find' command: /usr/bin/find
[12:36:59] Info: Found the 'ifconfig' command: /usr/sbin/ifconfig
[12:36:59] Info: Found the 'ip' command: /usr/sbin/ip
[12:36:59] Info: Found the 'ipcs' command: /usr/bin/ipcs
[12:36:59] Info: Found the 'ldd' command: /usr/bin/ldd
[12:36:59] Info: Found the 'lsattr' command: /usr/bin/lsattr
[12:36:59] Info: Found the 'lsmod' command: /usr/sbin/lsmod
[12:36:59] Info: Found the 'lsof' command: /usr/bin/lsof
[12:36:59] Info: Found the 'mktemp' command: /usr/bin/mktemp
[12:36:59] Info: Found the 'netstat' command: /usr/bin/netstat
[12:36:59] Info: Found the 'numfmt' command: /usr/bin/numfmt
[12:37:00] Info: Found the 'perl' command: /usr/bin/perl
[12:37:00] Info: Found the 'pgrep' command: /usr/bin/pgrep
[12:37:00] Info: Found the 'ps' command: /usr/bin/ps
:
```

Task 5 - Additional Detection Tools

1.Install andRunUnhide Using

commands : sudo apt install unhide -y sudo

unhide proc

sudo unhide sys

```
http://www.unhide-forensics.info

NOTE : This version of unhide is for systems using Linux >= 2.6

Used options:
[*]Searching for Hidden processes through /proc stat scanning

remnux@remnux:~$ sudo unhide sys
Unhide 20211016
Copyright © 2010-2021 Yago Jesus & Patrick Gouin
License GPLv3+ : GNU GPL version 3 or later
http://www.unhide-forensics.info

NOTE : This version of unhide is for systems using Linux >= 2.6

Used options:
[*]Searching for Hidden processes through getpriority() scanning
```

```
remnux@remnux:~$ sudo apt install unhide -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unhide is already the newest version (20220611-lubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 281 not upgraded.
remnux@remnux:~$ sudo unhide proc
Unhide 20211016
Copyright © 2010-2021 Yago Jesus & Patrick Gouin
License GPLv3+ : GNU GPL version 3 or later
http://www.unhide-forensics.info

NOTE : This version of unhide is for systems using Linux >= 2.6

Used options:
[*]Searching for Hidden processes through /proc stat scanning
```

2. Check for Kernel Modules

```
remnux@remnux:~$ lsmod | grep -i "hide\|rootkit"
remnux@remnux:~$
```

Results

1. Identified System Configuration

- Tools installed:
 - Rkhunter
 - unhide
- Baseline created using --propupd

2. Identified Scan Results

- Most checks passed successfully •Some warnings detected: Hidden directories◦
 - Modified configuration files
- No confirmed rootkit infection

Using Rkhunter, the system was examined to detect rootkits and unusual changes. Although a few alerts were generated, they were determined to be normal system activity or false positives. Further verification with tools like *unhide* and *lsmod* showed no evidence of hidden processes or malicious kernel modules. This highlights the need to use multiple tools to ensure reliable and accurate rootkit detection.